

61. Option (a) is correct.

We have two sides of square lie on the lines

$$2x + y - 3 = 0 \quad \dots(i)$$

$$4x + 2y + 5 = 0 \quad \dots(ii)$$

Divide (ii) by 2 we get

$$2x + y + \frac{5}{2} = 0 \quad \dots(iii)$$

Here $2x + y - 3 = 0$ and $2x + y + \frac{5}{2} = 0$ are

parallel lines.

Therefore, length of the side of the square =

Distance between parallel side

$$= \frac{|c_1 - c_2|}{\sqrt{a_1^2 + b_1^2}}$$

Where $c_1 = -3$ and $c_2 = \frac{5}{2}$

$$\begin{aligned} (a_1, b_1) &= (2, 1) \\ \Rightarrow \frac{\left| -3 - \frac{5}{2} \right|}{\sqrt{(2)^2 + (1)^2}} &= \frac{\left| -\frac{11}{2} \right|}{\sqrt{5}} \\ &= \frac{11}{2\sqrt{5}} \end{aligned}$$

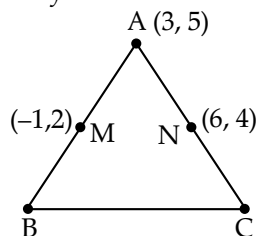
Length of the side of square = $\frac{11}{2\sqrt{5}}$ units

$$\begin{aligned} \text{Area of the square} &= (\text{side})^2 = \left(\frac{11}{2\sqrt{5}} \right)^2 \\ &= 6.05 \text{ square units.} \end{aligned}$$

62. Option (b) is correct.

Given $A = (3, 5)$ and

Mid-point of sides AB and AC is $(-1, 2)$ and $(6, 4)$ respectively.



Let $B = (x_1, y_1)$ and $C = (x_2, y_2)$

By using mid-point formula we will find the point B and C

$$\begin{aligned} M &= (-1, 2) \\ &= \left(\frac{x_1 + 3}{2}, \frac{y_1 + 5}{2} \right) \end{aligned}$$

$$N = (6, 4)$$

$$= \left(\frac{x_2 + 3}{2}, \frac{y_2 + 5}{2} \right)$$

By comparing we get

$$\frac{x_1 + 3}{2} = -1, \quad \frac{y_1 + 5}{2} = 2$$

$$\Rightarrow x_1 = -5, y_1 = -1$$

$$\text{and } \frac{x_2 + 3}{2} = 6, \quad \frac{y_2 + 5}{2} = 4$$

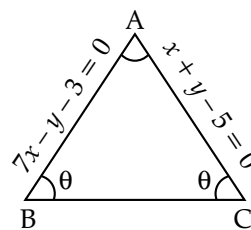
$$\Rightarrow x_2 = 9, y_2 = 3$$

$$\therefore B = (-5, -1) \text{ and } C(9, 3)$$

$$\begin{aligned} \text{Centroid of } \triangle ABC &= \left(\frac{3 - 5 + 9}{3}, \frac{5 - 1 + 3}{3} \right) \\ &= \left(\frac{7}{3}, \frac{7}{3} \right) \end{aligned}$$

63. Option (b) is correct.

Given that ABC is an acute angled isosceles triangle



First we find slope of $AB(m_1) = 7$

and slope of $AC(m_2) = -1$

$$\text{We know that } \tan A = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{7 - (-1)}{1 + (-7)} \right|$$

$$= \frac{8}{6} = \frac{4}{3}$$

$$\Rightarrow \tan A = \frac{4}{3}$$

$$A = \tan^{-1} \frac{4}{3}$$

We know that sum of all the angle of triangle = π

$$\text{i.e., } \theta + \theta + \tan^{-1} \frac{4}{3} = \pi$$

$$\Rightarrow 2\theta = \pi - \tan^{-1} \frac{4}{3}$$

$$\Rightarrow \theta = \frac{\pi}{2} - \frac{1}{2} \tan^{-1} \frac{4}{3}$$